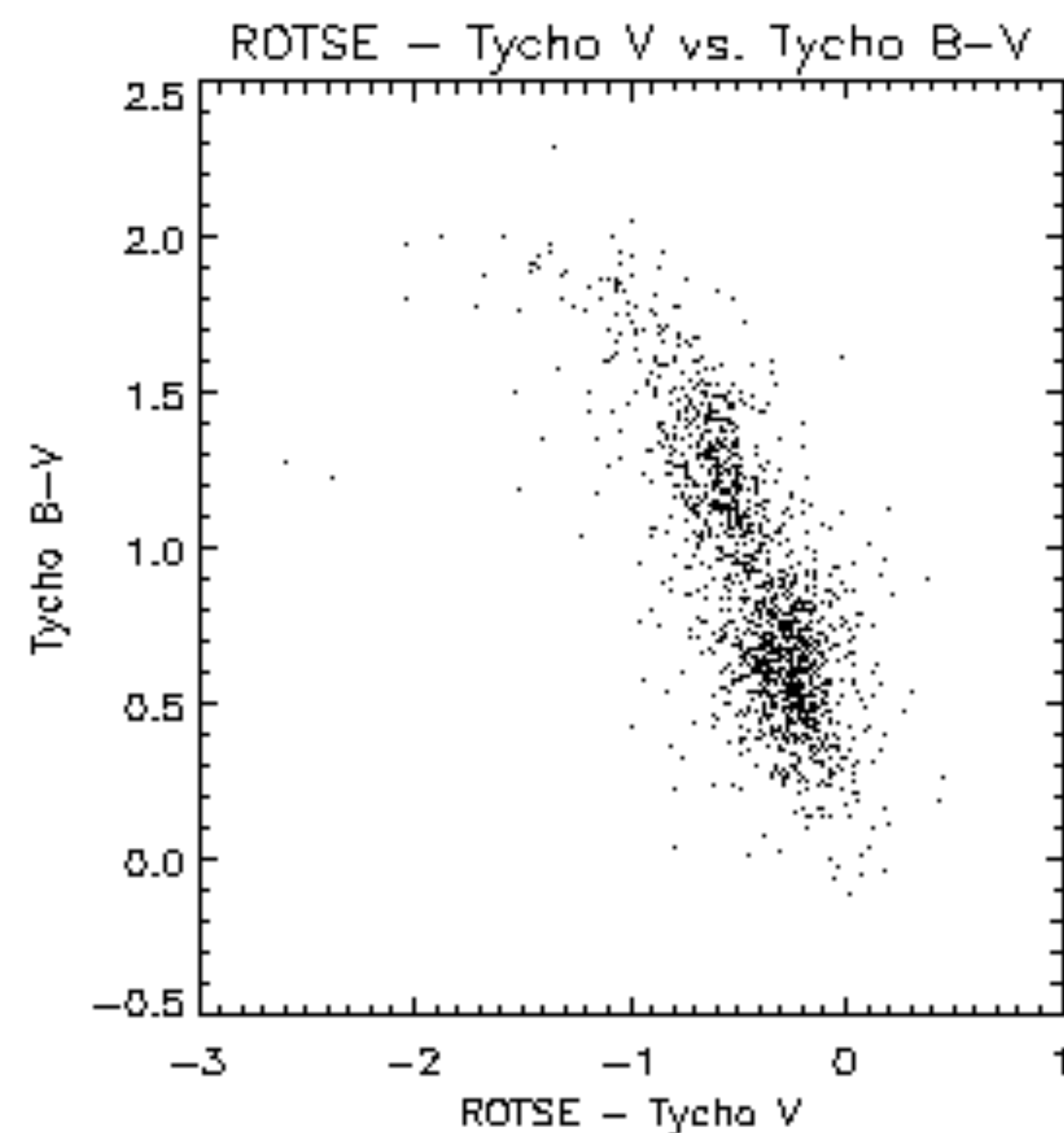
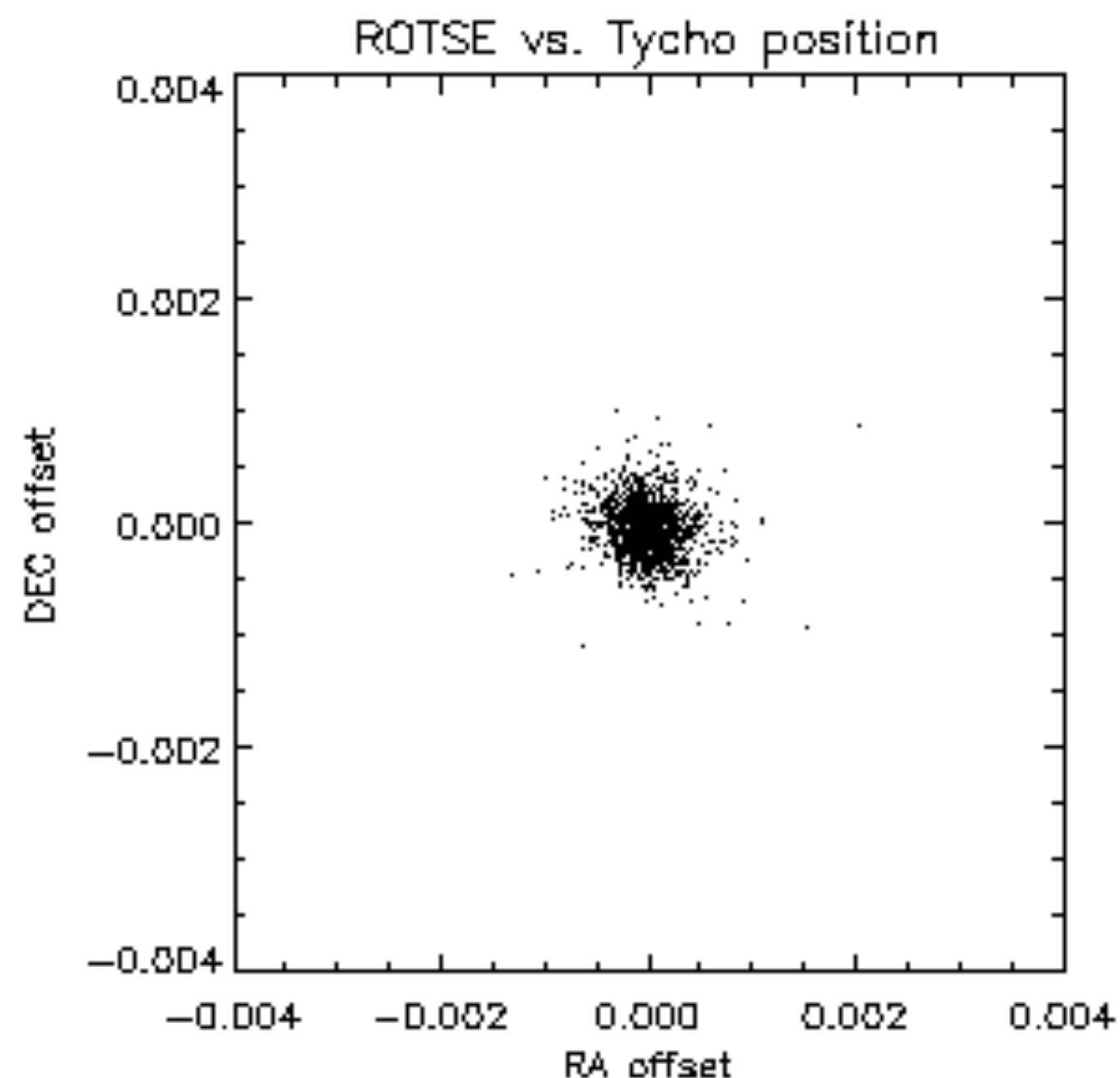




- ▶ IDL Basics
 - ▶ Selecting candidate lightcurves
 - ▶ Statistical concepts
 - ▶ ID OBJECTs of interest
- ▶ SN photometry
- ▶ Telescope performance
 - ▶ CCDs, astrometry, photometry

LAB 1

LIGHTCURVES

CONVERTING BETWEEN FLUX AND MAGNITUDE

- ▶ Fundamental physics quantities
 - ▶ Flux in units of energy/area/time
- ▶ Often measurements are in magnitudes and easier to handle that way
- ▶ Flux to magnitude conversion

$$m - m_0 = -2.5 \log \left(\frac{F}{F_0} \right)$$

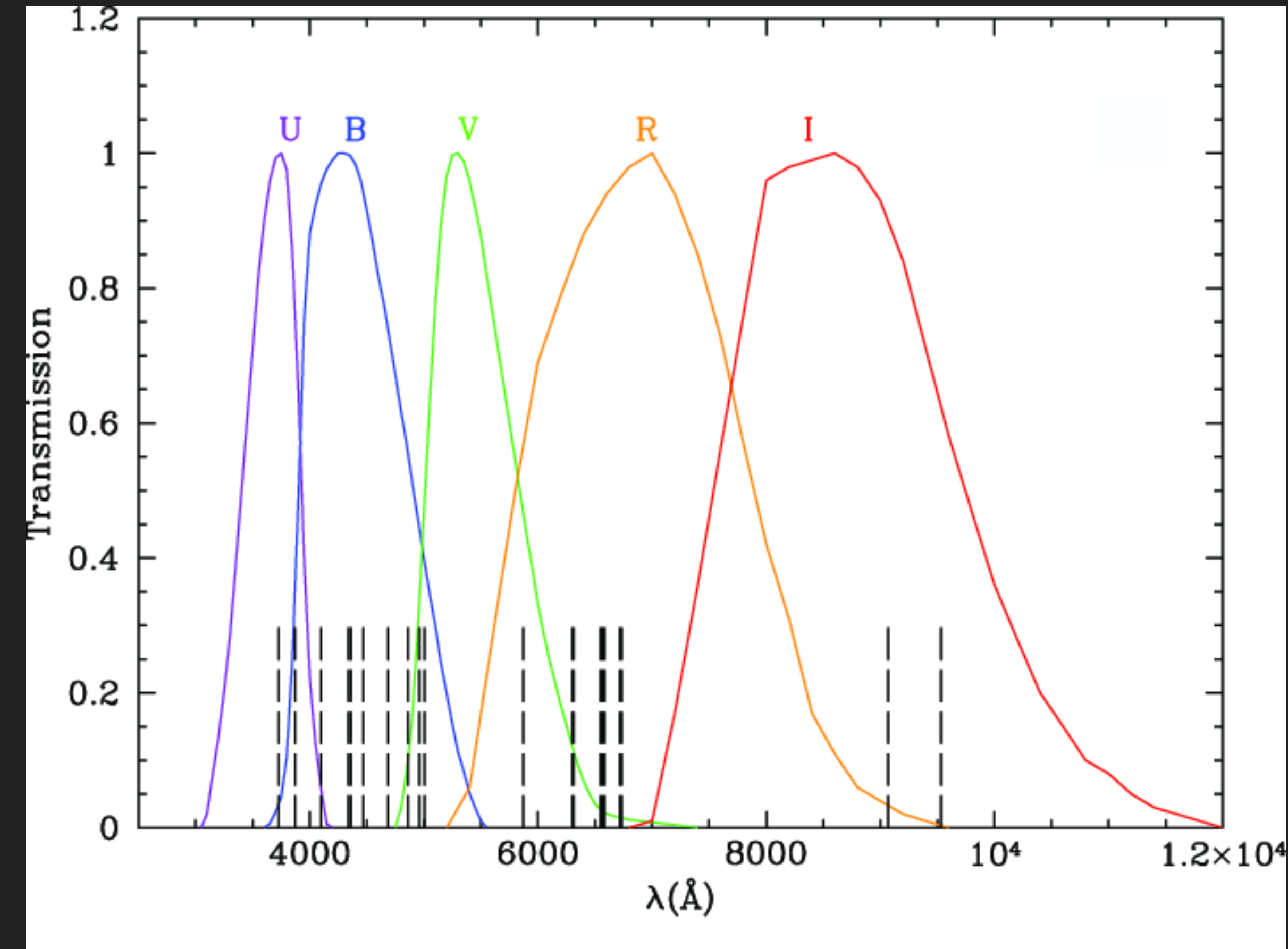
- ▶ Since magnitude logarithmic, a difference corresponds to a ratio of luminosities
 - ▶ Need a reference, F_0 , of the flux that corresponds to m_0 , generally close to 0
 - ▶ called the 'zeropoint' of the magnitude scale
 - ▶ Each photometric band has a different zeropoint associated with it
- ▶ Magnitude to flux conversion:

$$F = F_0 \cdot 10^{-0.4 \Delta m}$$

Johnson Filter System:		Vega
F(V) (10 ⁻⁹ erg/s/cm ² /Å)		3.55
F(B) (10 ⁻⁹ erg/s/cm ² /Å)		6.20
Fluxes from: Colina, Bohlin, "Absolute Flux calibration of Optical Spectrophotometric Standard Stars," Astron. Journal 108:1931 (1994)		
V		0.044
B		0.163

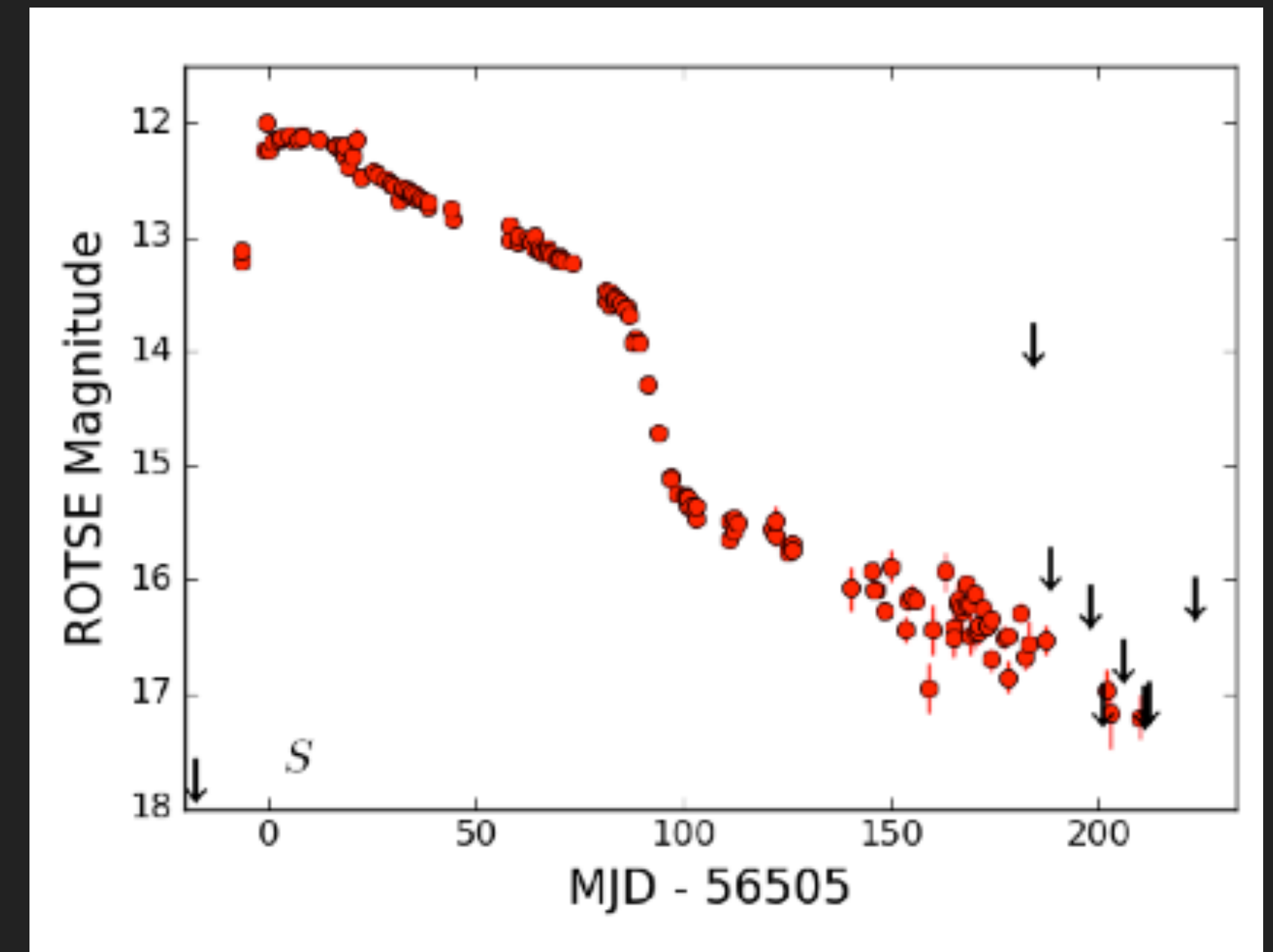
PHOTOMETRIC BANDS

- ▶ Narrow bands at UV, optical or infrared wavelengths
 - ▶ Obtain magnitudes in these bands
- ▶ Several systems
 - ▶ Johnsons-Cousins: *UBVRI*
 - ▶ Gunn: *ugriz*



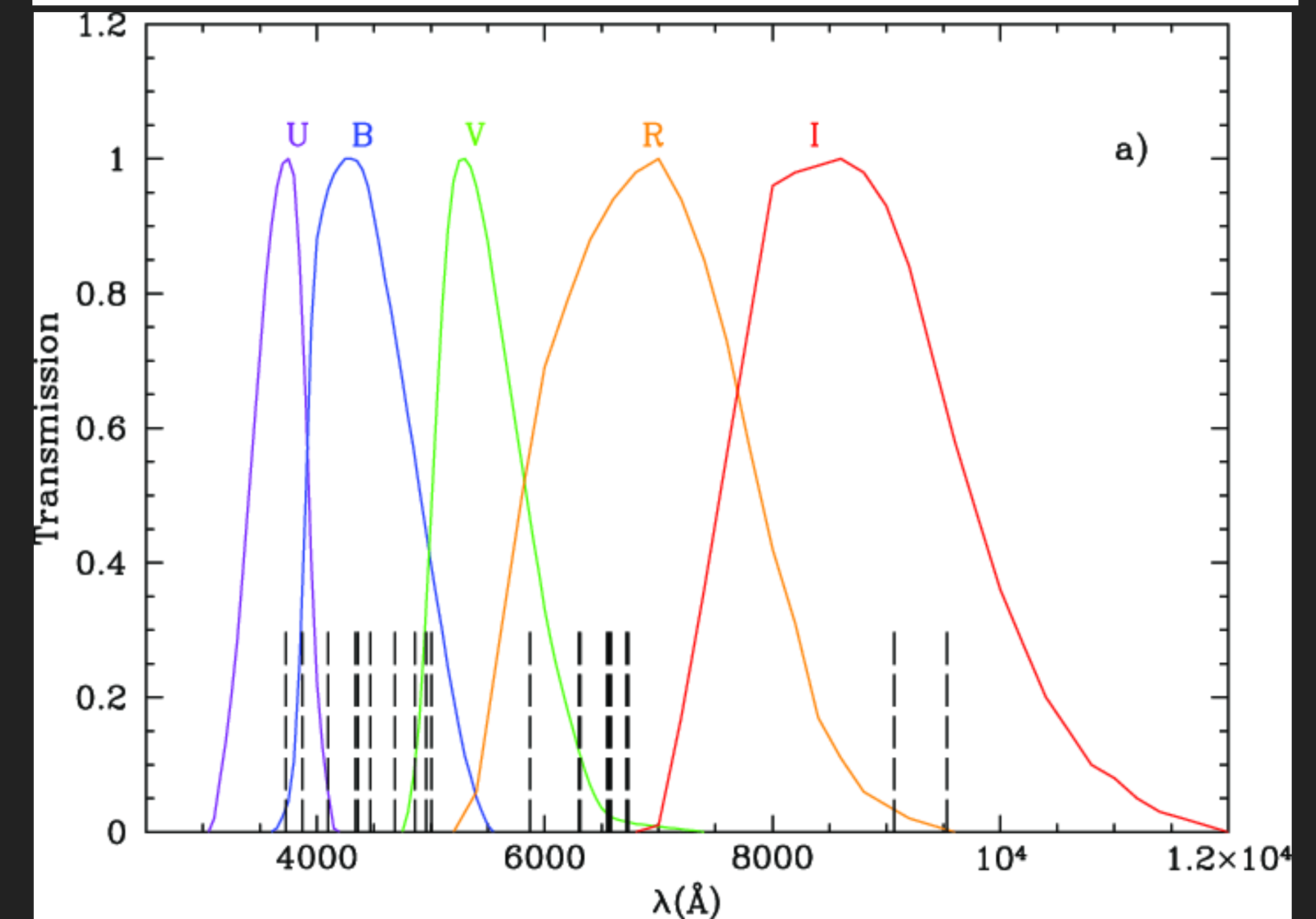
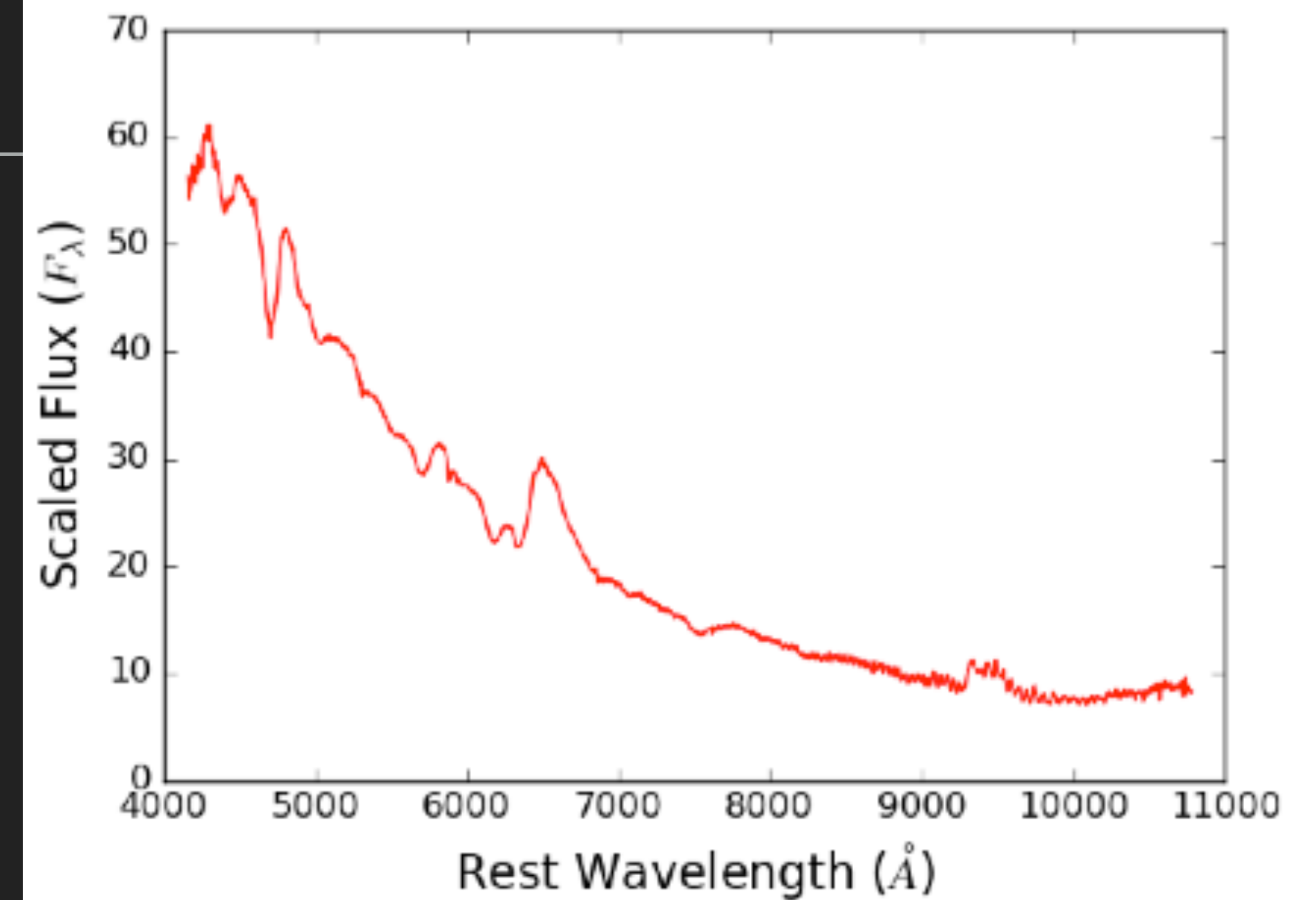
SHOWING THE LIGHTCURVE

- ▶ Lightcurve data in snXph file(s)
 - ▶ Will generally correspond to magnitude with no filter
 - ▶ V passband
 - ▶ Need to convert from flux units/angstrom to magnitude
 - ▶ Do your V-band magnitudes agree roughly with the photometry?
- ▶ Calculate color index
 - ▶ B-V
 - ▶ At epoch you have spectrum



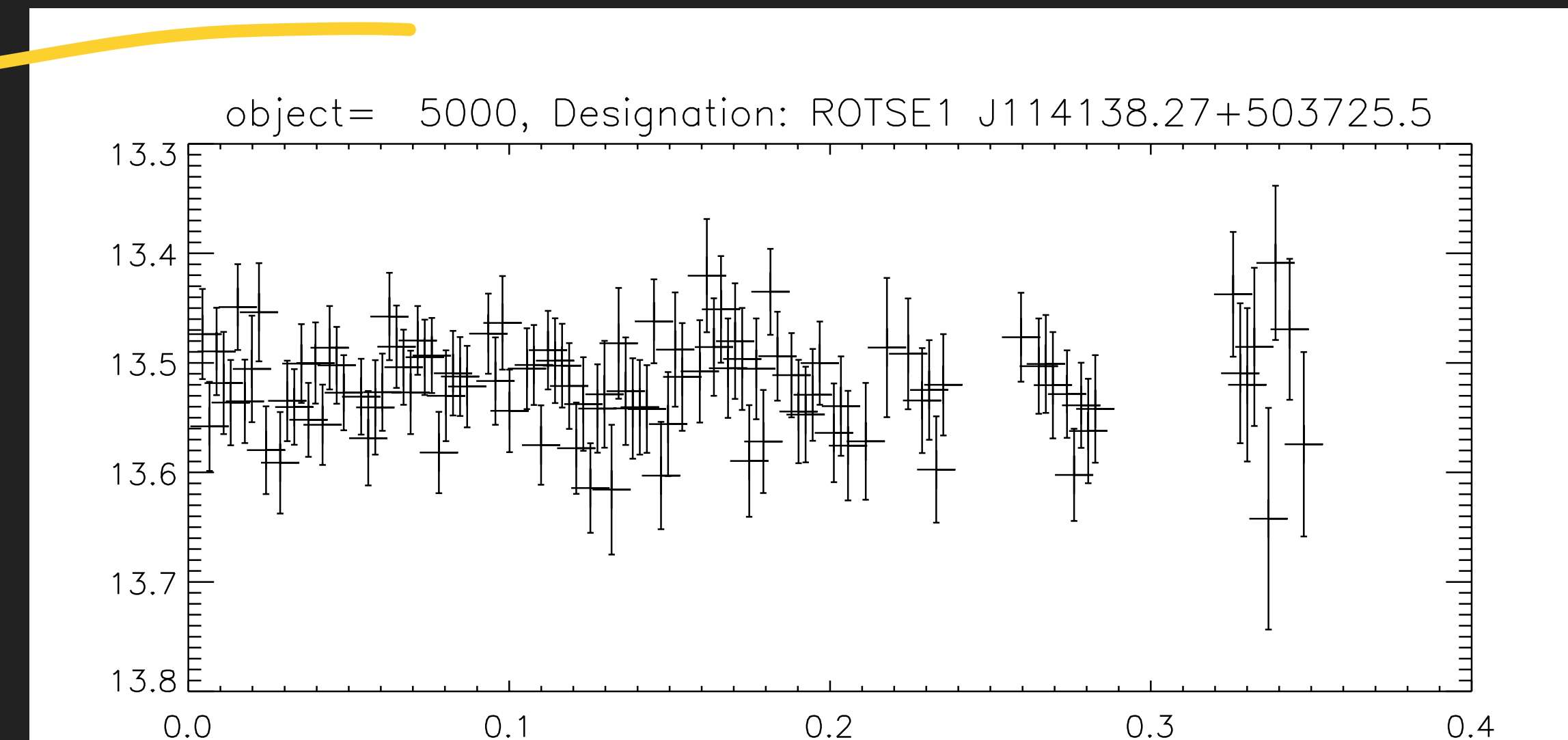
BROADBAND MAGNITUDE AND TEMPERATURE

- ▶ Integrate spectrum over broadband
 - ▶ B passband
 - ▶ V passband
 - ▶ Need to convert from flux units/angstrom to magnitude
 - ▶ Do your V-band magnitudes agree roughly with the photometry?
- ▶ Calculate color index
 - ▶ B-V
 - ▶ At epoch you have spectrum



LIGHTCURVES

- ▶ Now select an object number – it doesn't matter which one for now. I'll pick 5000.
- ▶ Run a procedure 'find_burst' to extract the data and plot a lightcurve for this object
- ▶ IDL> `x=find_burst(match, 0,0,objid=5000,log='firsttry')`
- ▶ IDL> `$ls *.ps`
- ▶ IDL> `$evince firsttry.ps &`
- ▶ Let's have a look at your plot!



LAB 2: FINDING VARIABLE STAR CANDIDATES

SOURCE SELECTION

- ▶ Picking up where we left off, let's examine our match structure further to select lightcurves that appear to be varying

- ▶ We use 'find_burst', but this time we will use it to select objects that are statistically different than constant

- ▶ 3 statistical quantities we can cut on to select variable candidates

- ▶ **delta_magnitude**: the difference in magnitude between the brightest and dimmest measurement in a lightcurve: $\Delta m = m_{max} - m_{min}$

- ▶ **Maximum significance**: the significance of the maximum variation: $\sigma_{max} = \frac{(m_{max} - m_{min})}{\sqrt{\epsilon_{max}^2 + \epsilon_{min}^2}}$

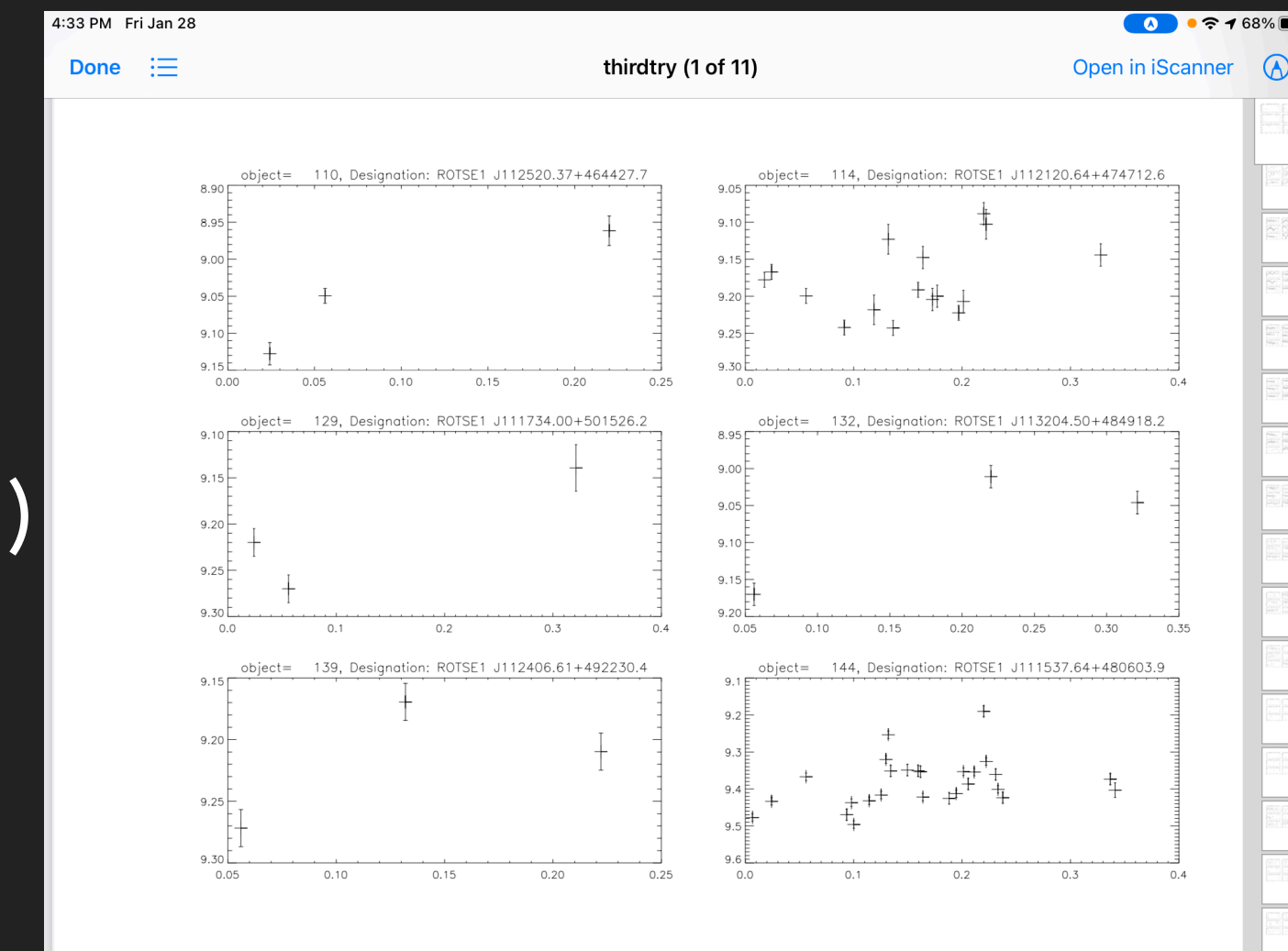
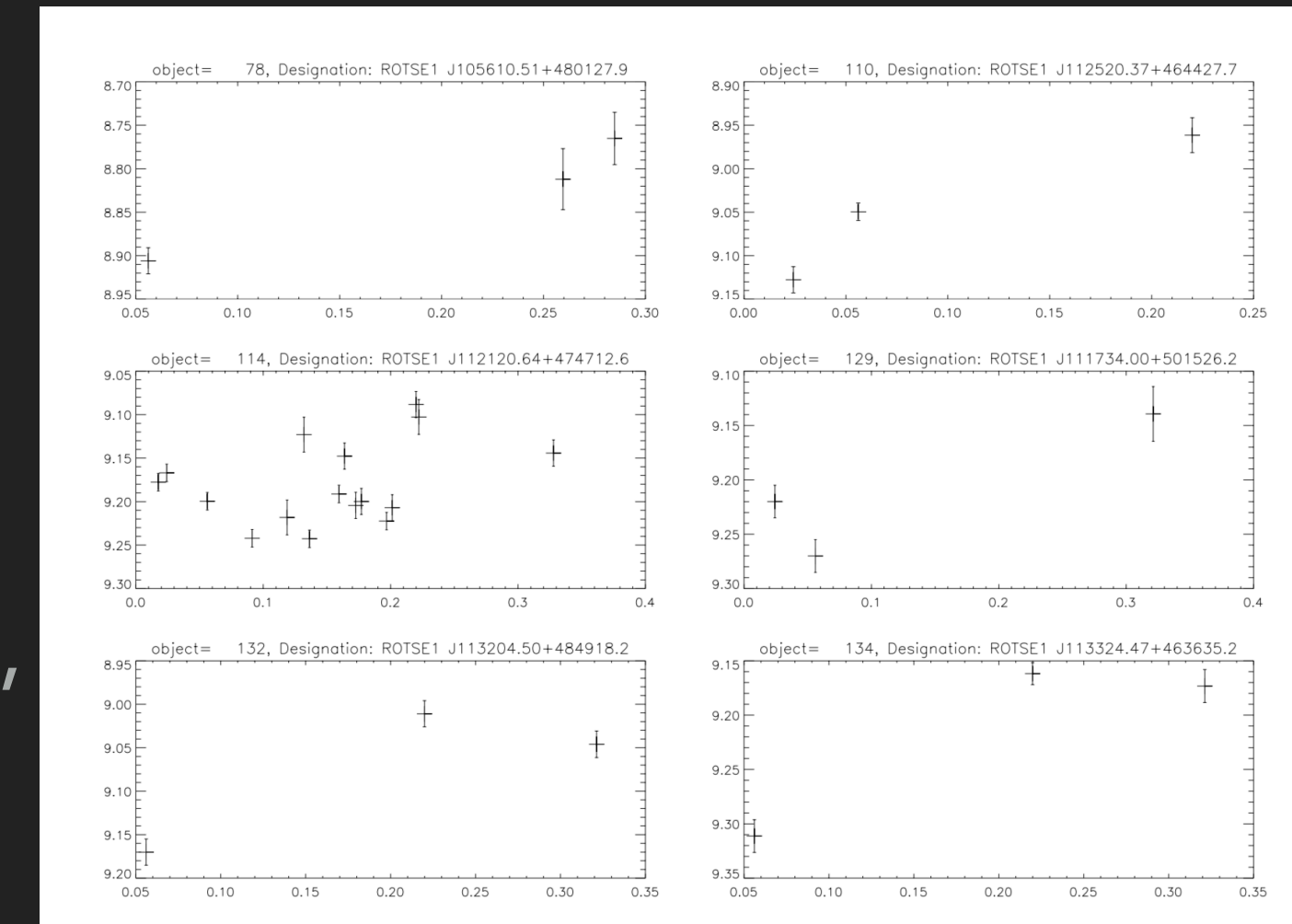
- ▶ **Minimum chi-squared**: the square of the difference of each measurement from the median of a lightcurve, divided by the square of the statistical uncertainty on this difference:

$$\chi^2 = \sum \left[\frac{(m_i - m_{avg})}{\sigma_i} \right]^2$$

- ▶ Values are summed up over all measurements in a lightcurve and divided by the number of measurements minus 2, to give a 'χ² per degree of freedom'
- ▶ You can find more about these in the VSP paper: E. Fagg, et al., Journal of Undergraduate Research in Physics (2009).

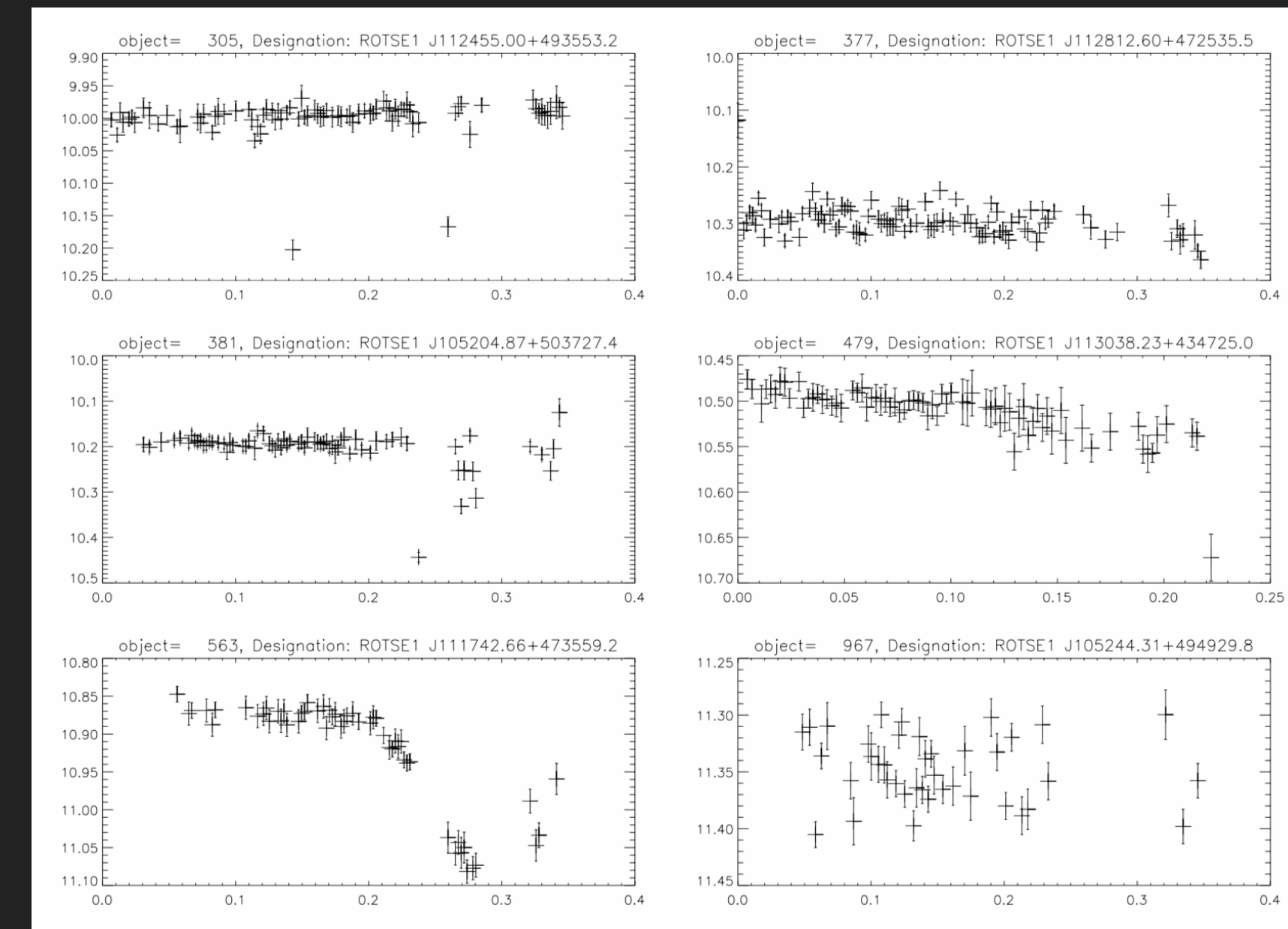
FINDING CANDIDATE VARIABLES

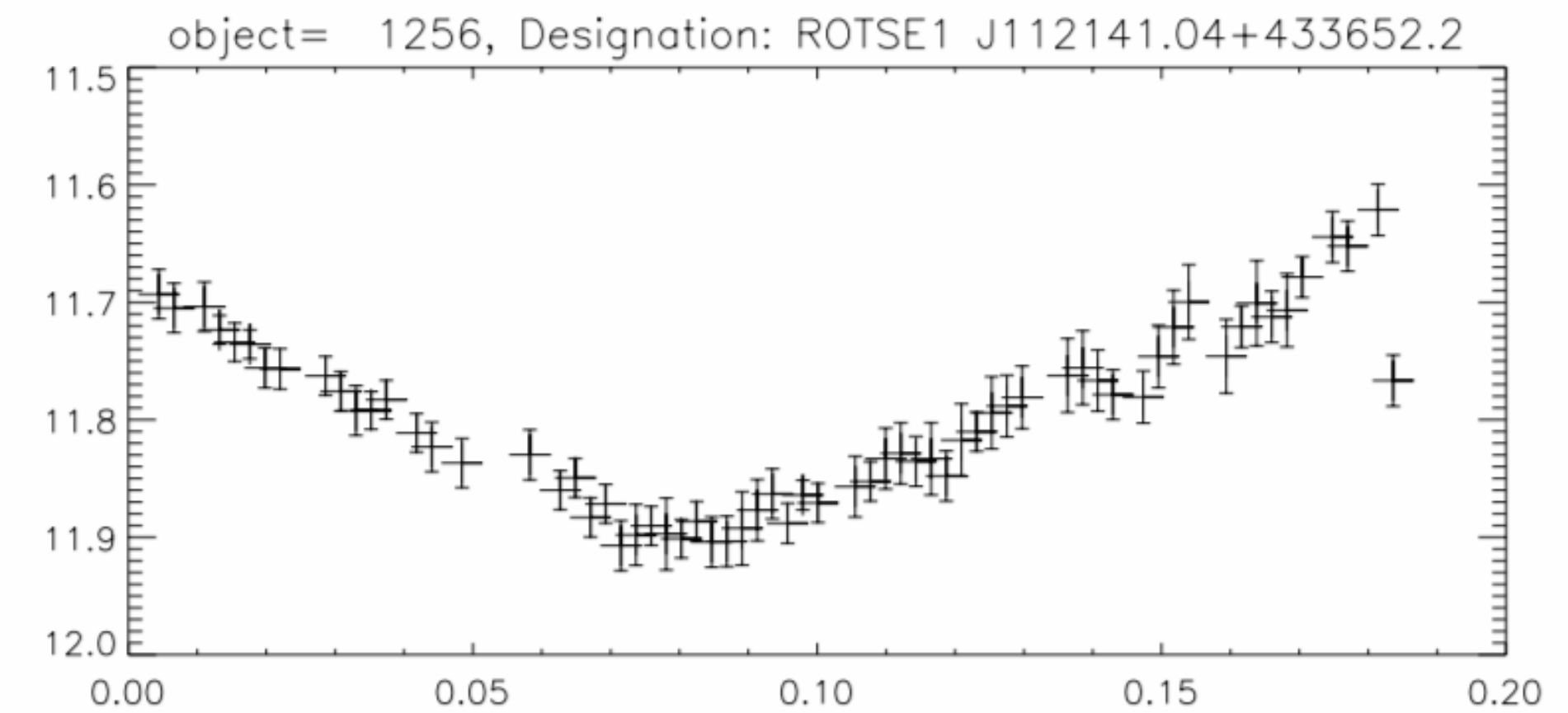
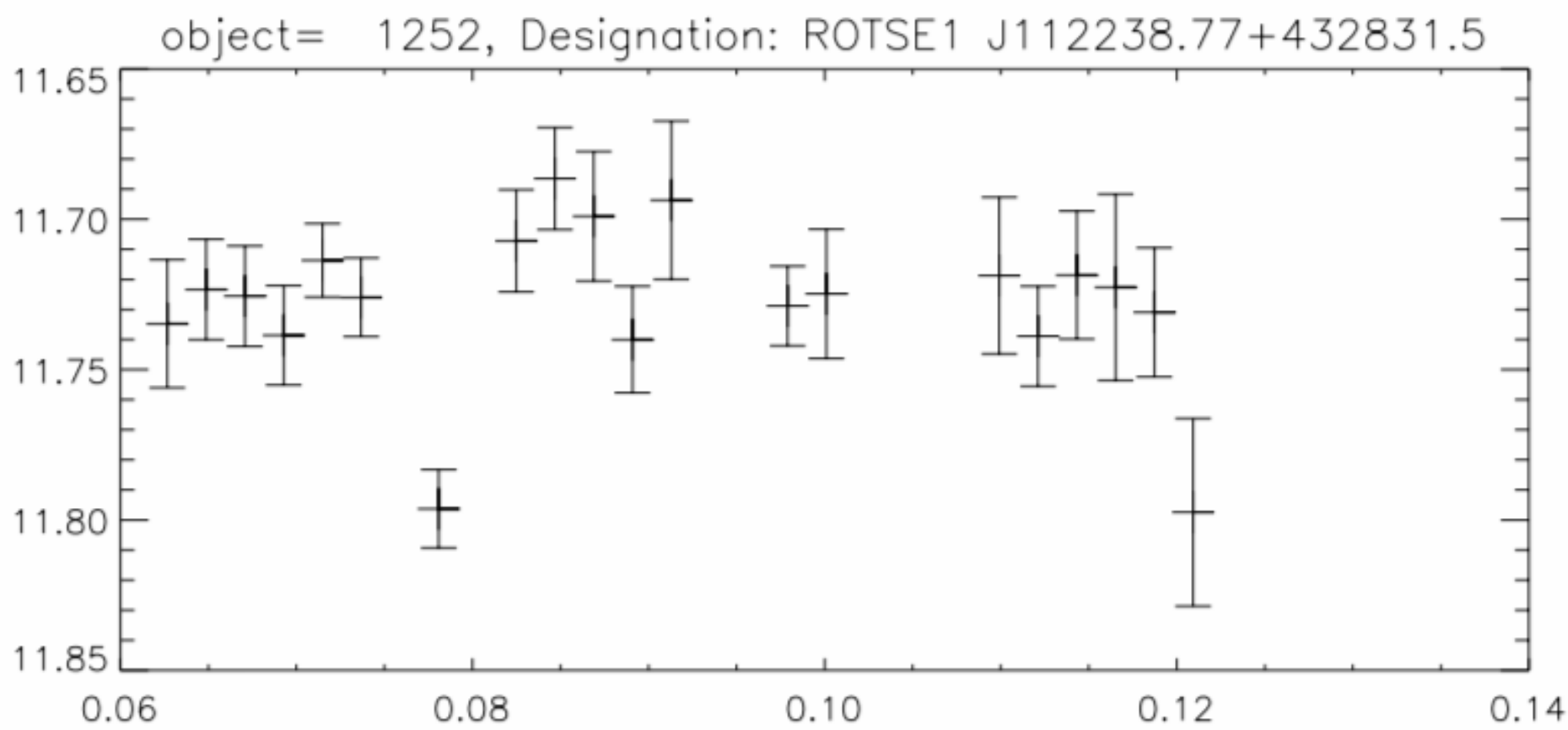
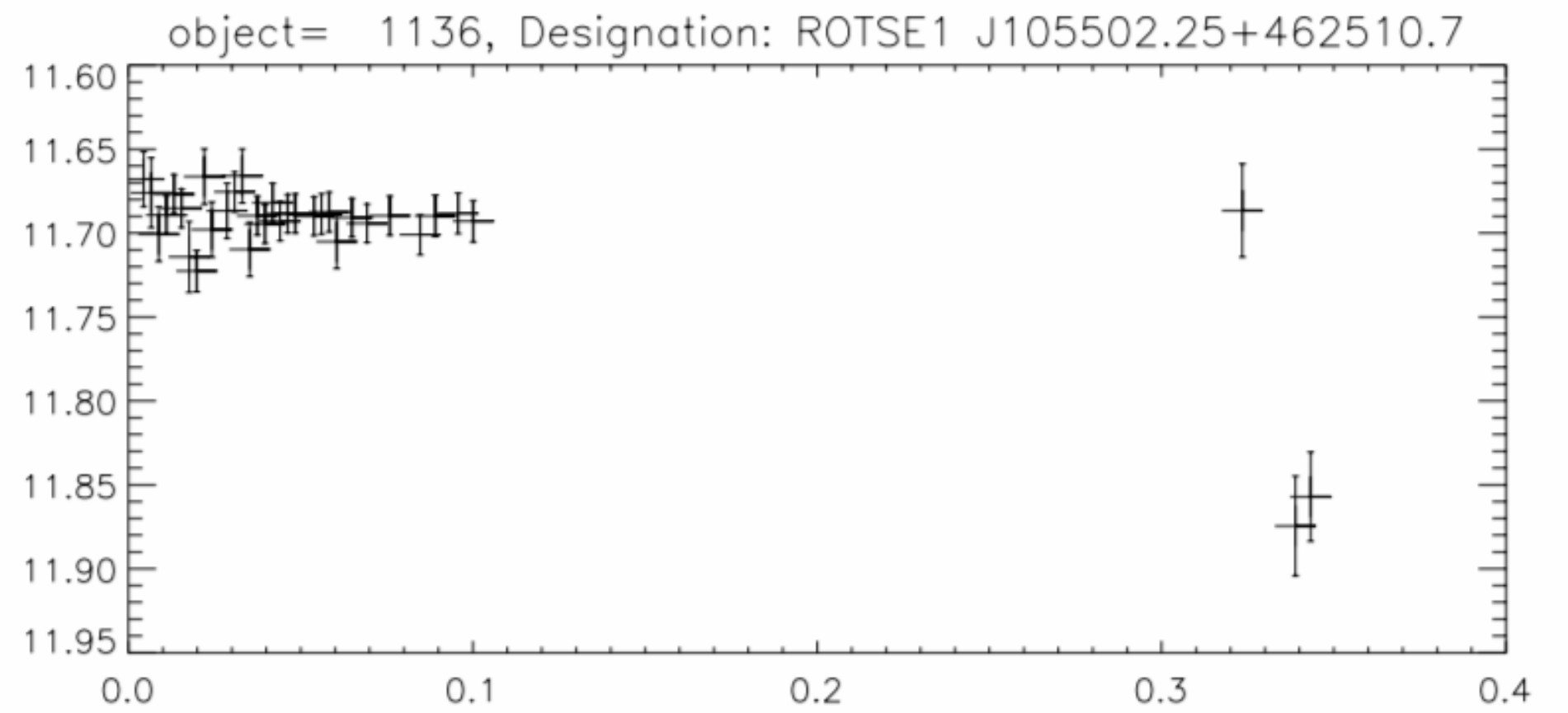
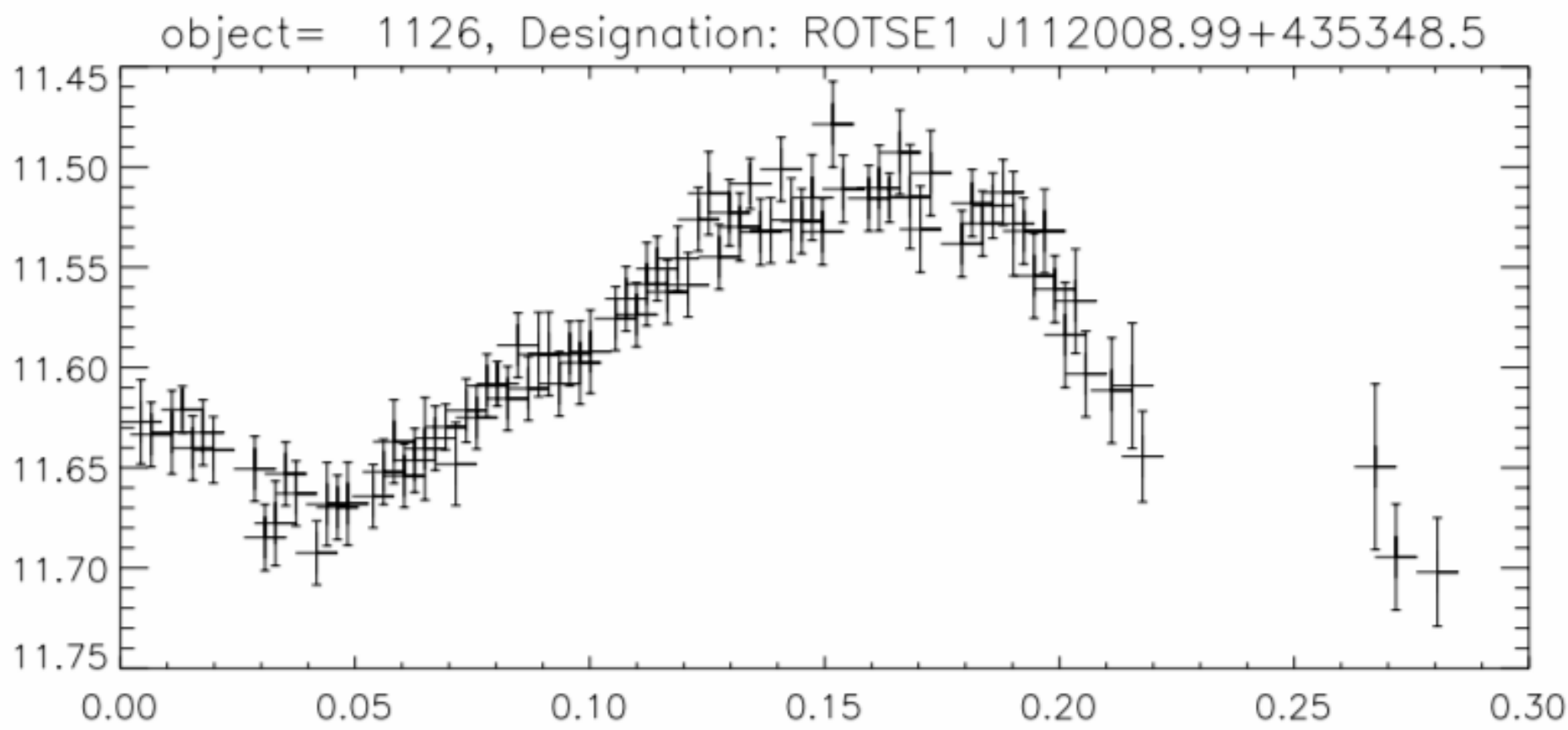
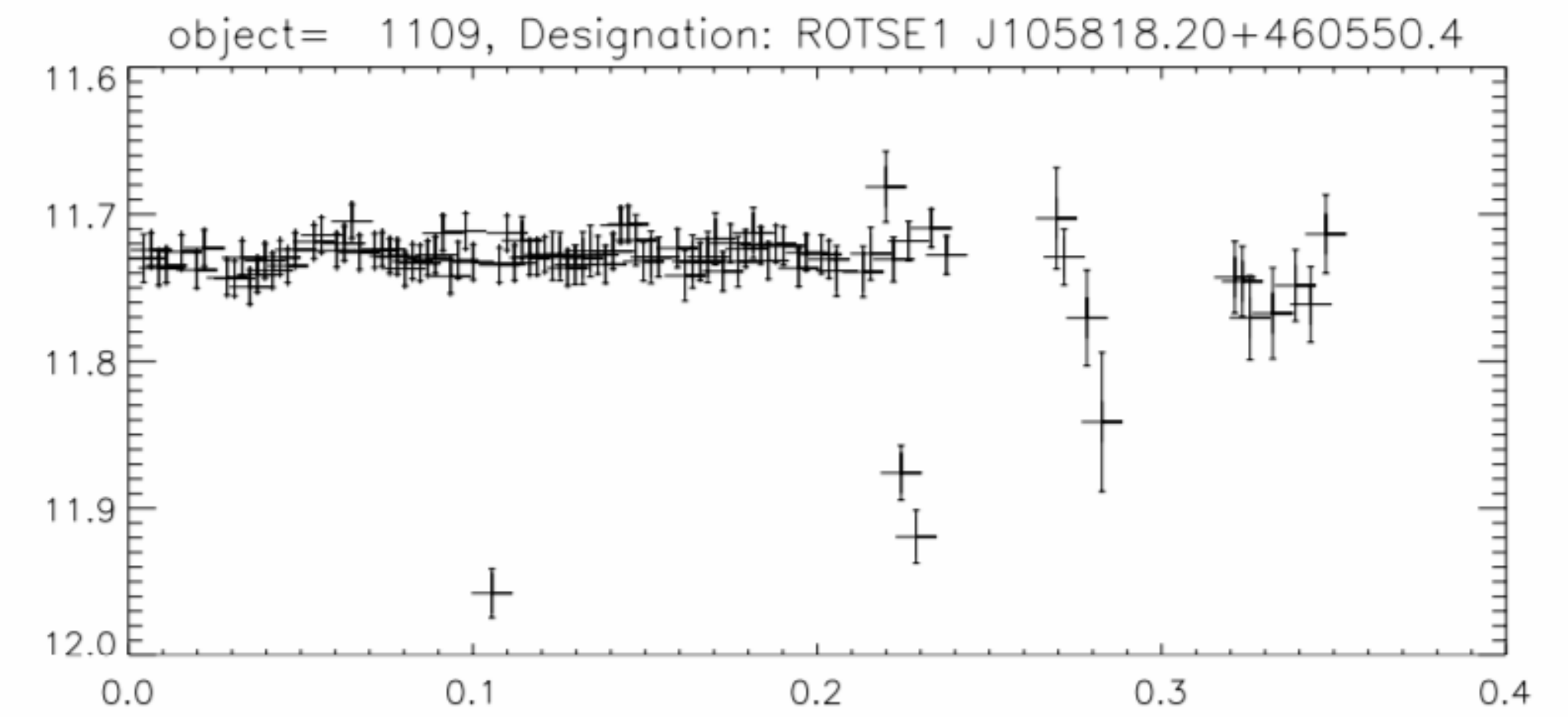
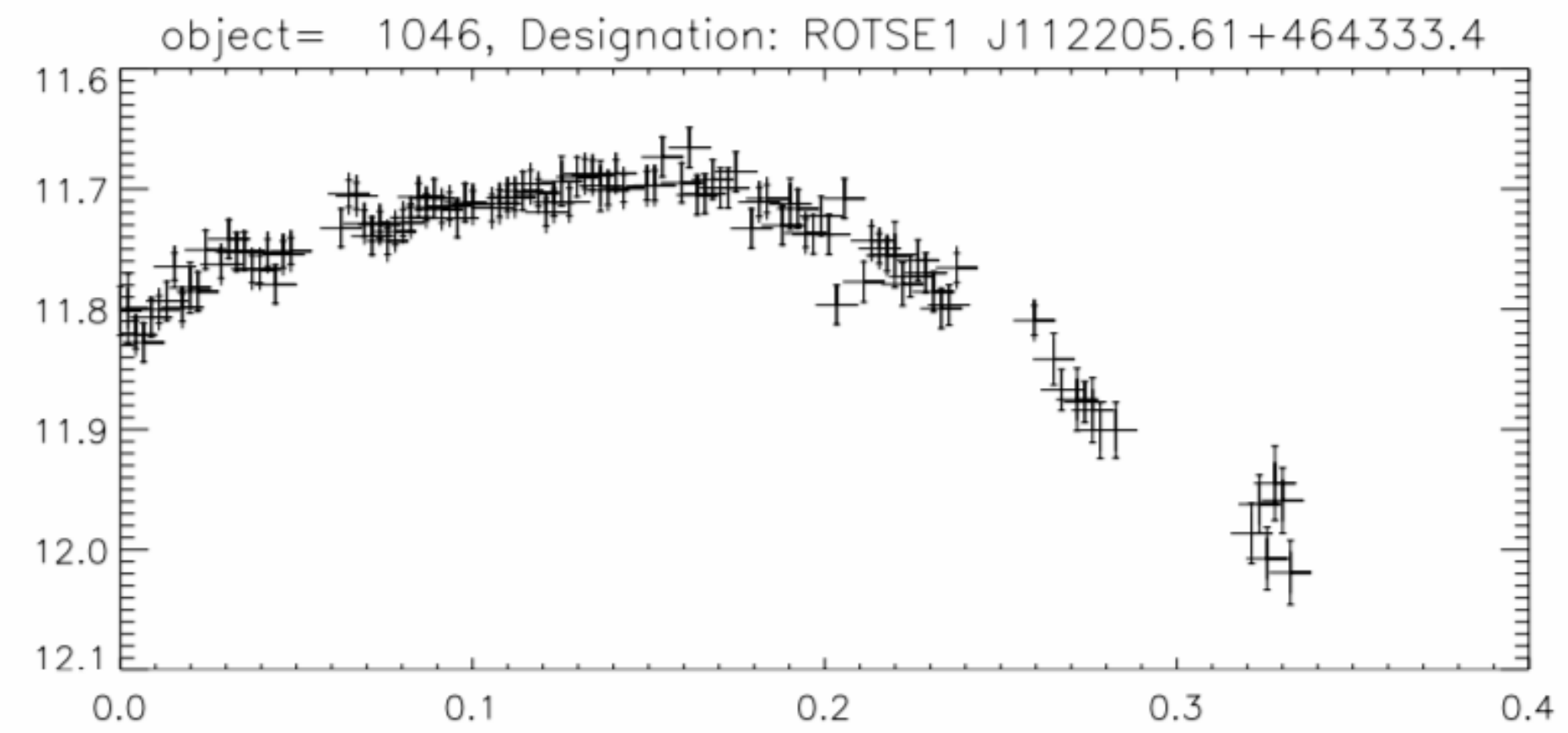
- ▶ How do we use find_burst in this way?
 - ▶ IDL> x=find_burst(match, 0.1,3.0,log='secondtry')
 - ▶ Where 2nd and 3rd arguments are delta_mag and max_sig, respectively
 - ▶ Examine outputs to see what you get:
 - ▶ IDL> \$evince secondtry.ps &
 - ▶ Examine this file, in all it's pages.
 - ▶ How many objects are found?
 - ▶ What happens when you vary delta_mag? Max_sig?
 - ▶ Now let's add a powerful variable, chi-squared:
 - ▶ IDL> x=find_burst(match, 0.1,3.0,minchisq=2.0,log='thirdtry')
 - ▶ What happens to the # of candidates now?



FINDING CANDIDATE VARIABLES

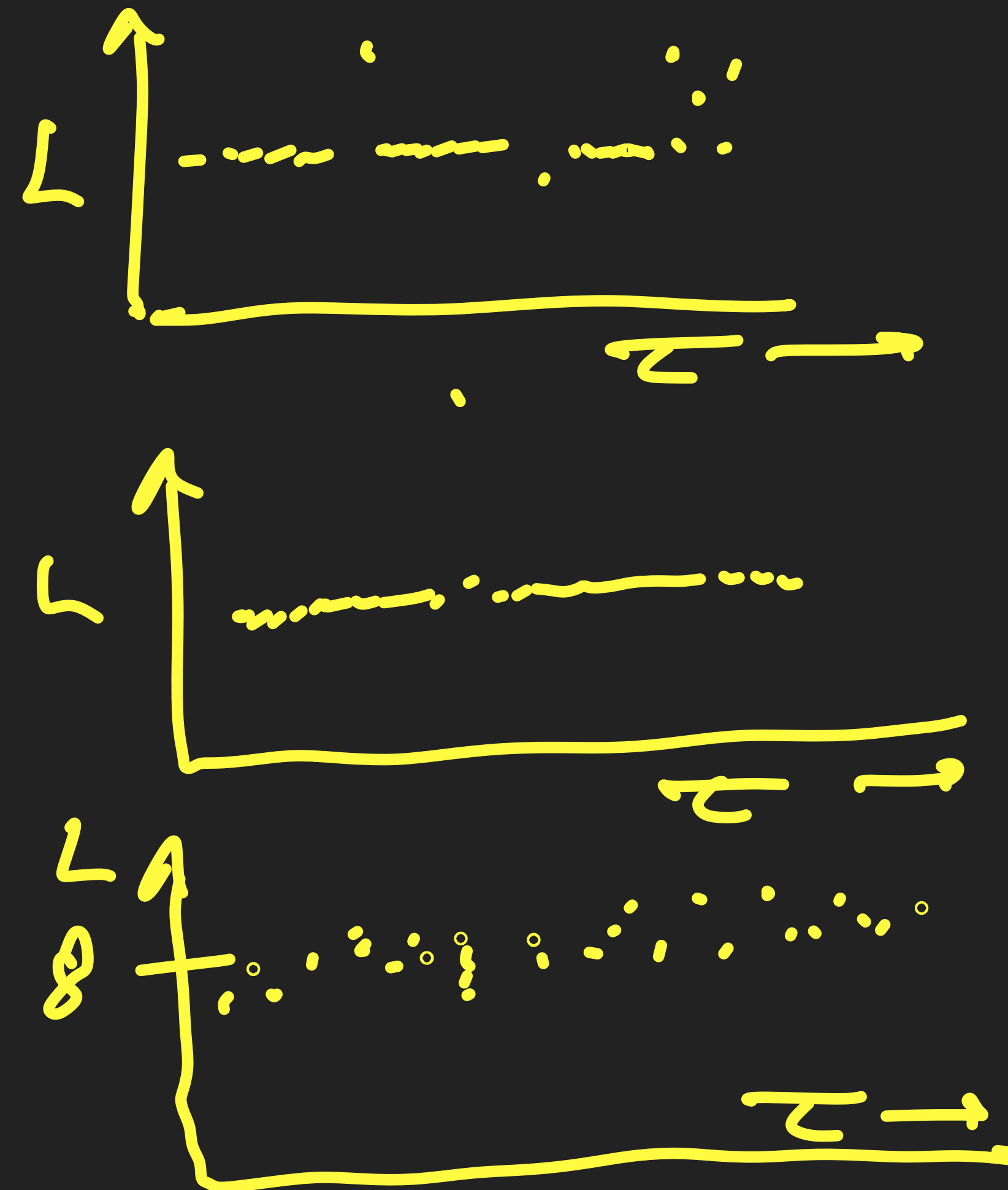
- ▶ Now, look at the individual lightcurves for all pages in your .ps files
- ▶ Can you pick out lightcurves that are physical, and those that may be unphysical?
 - ▶ Why are there lightcurves that are not physical?
 - ▶ What is happening there?
 - ▶ What physics could produce the physical lightcurves?
- ▶ You will notice a lot of 'background' from apparent variations that have nothing to do with variables
 - ▶ How do we handle these?
 - ▶ We will revisit this next time.





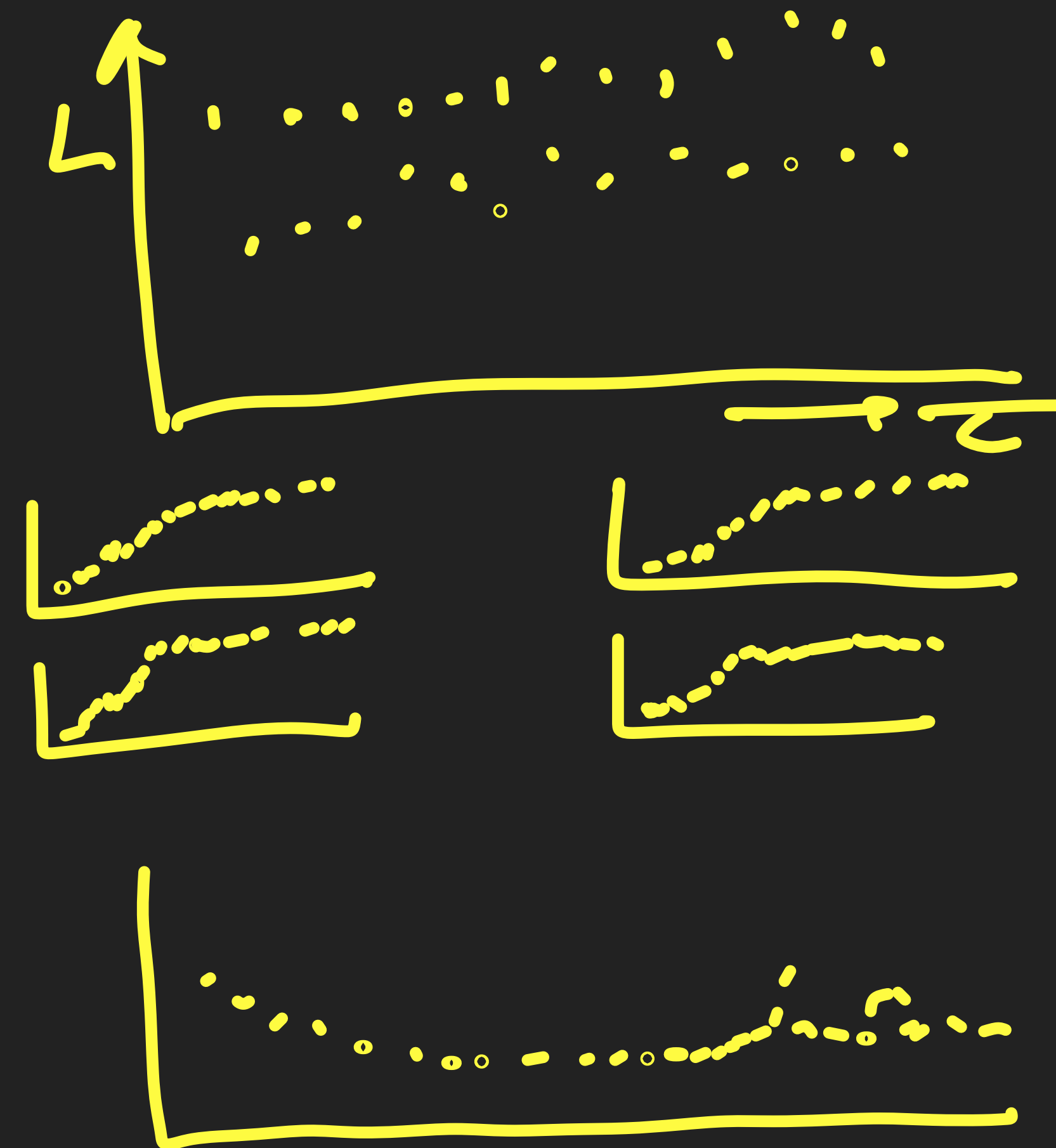
INSTRUMENTAL CANDIDATES IN LIGHTCURVES

- ▶ **Hot pixel** - occasional, erratic change in brightness in lone exposures
- ▶ **Cosmic rays** - single brightened observation
- ▶ **Saturation** - all bright stars seem variable (mag < 9-10)



INSTRUMENTAL CANDIDATES IN LIGHTCURVES (CONT)

- ▶ **Blending** - alternating magnitudes statistically DES repaint from each other
- ▶ **Weather** vs. time - slow trend over night, but for several nearby objects in a camera field
- ▶ **Vignetting** vs. time - possibly slow trend or intermittent ('sticking') behavior in corners of camera field



SUPPORTING MATERIAL

- ▶ There is a 'manual' written by former students using this data for research:
 - ▶ https://docs.google.com/document/d/1vEw-jgaDF4lw98pGBkhn5T5LEG255ynzC_aWDrKADUY
 - ▶ The first 3 chapters are useful for this course
- ▶ Also the web-site for ROTSE, rotse.net, has several references that may be of use at:
 - ▶ <http://rotse.net/publications/publications.html>
 - ▶ In particular, Kehoe et al. 2002 (source of the data for this course).